MCO-1D Electronics Bill of Materials

Metachronal Cascade Ornithopter — Drone Prototype · USD Pricing with Amazon.com Links · Rev H Specification · 2026-06-03

How to use this BOM: Every Amazon link is a search or product page — verify the exact item matches the specification before purchasing. Prices are approximate at mid-2026. The ODrive Micro is NOT sold on Amazon — order direct from odriverobotics.com. Items marked 'custom' require 3D printing or machining. Airframe, wing materials, tools, and LiPo charger are not included.

FLAP DRIVE

Item	Spec	Qty	Unit	Line	Notes / Amazon Link
GoolRC 5010 360KV Brushless Motor	5010 frame · 360KV · ~90g · 2–6S	8	$15.99	$127.92	Confirmed Rev H. One per wing. Search 'GoolRC 5010 360KV' for best price — multiple sellers from ~$13. Buy pack of 4 if available. https://www.amazon.com/dp/B0C2YRNX1Z
ODrive Micro motor controller	ODrive Micro · 6S · CAN-FD · hall encoder	8	$79.00	$632.00	NOT on Amazon — order direct from odriverobotics.com. $79 each. The single biggest cost item. No substitute for CAN-FD closed-loop control. https://shop.odriverobotics.com/products/odrive-micro
Dyneema rope 1.8mm — 5m reel	UHMWPE braid · 1.8mm · ~880 lb break · SK75/SK99	2	$9.99	$19.98	~1m per wing = 8m total. Buy 2 reels. Search 'Dyneema 80lb braided fishing line'. https://www.amazon.com/s?k=dyneema+1.8mm+fishing+line
Drive pulley 12mm OD (motor shaft)	Custom · 12mm OD · 3mm bore · rope groove	8	$3.00	$24.00	Print in PETG for prototype. Search 'GT2 pulley 12T 3mm bore' as starting point — may need lathe modification. ~$3 estimate for custom print/modify. https://www.amazon.com/s?k=timing+pulley+12mm+3mm+bore
Driven pulley 84mm OD (wing crank)	Custom · 84mm OD · rope groove · 7:1 ratio	8	$5.00	$40.00	Print in PETG. 84÷12 = 7:1. No standard product — custom print or machine. ~$5 print cost. https://www.amazon.com/s?k=large+pulley+84mm

Subtotal: $843.90

SERVOS

Item	Spec	Qty	Unit	Line	Notes / Amazon Link
DS3225 25KG Digital Servo 270°	DS3225 · 25KG · 270° range · 60g · waterproof	8	$13.99	$111.92	Sweep axis. 270° range ESSENTIAL — must reach 210° stow. Verify listing specifies 270° (not 180°). ZOSKAY brand preferred — also search 'DS3225 270 degree'. https://www.amazon.com/dp/B07RNFQYD2
Emax ES08MD II Digital Micro Servo	ES08MD II · 12g · digital · metal gear · 4.8–6V	8	$7.99	$63.92	Pitch axis. Confirmed Rev H. 8 required. Also available in 4-packs for better value. https://www.amazon.com/dp/B0BF8WXYZM

Subtotal: $175.84

COMPUTERS

Item	Spec	Qty	Unit	Line	Notes / Amazon Link
Raspberry Pi Zero 2 W	RPi Zero 2W · quad-core 1GHz · 512MB · WiFi · 40-pin GPIO	1	$15.00	$15.00	Wave library CPU. Official price $15 but often marked up on Amazon. Check raspberrypi.com for authorized resellers. Stock can be limited. https://www.amazon.com/dp/B09NDBJ9QM
Pixhawk 6C Mini Flight Controller	STM32H743 · dual IMU · baro · PX4/ArduPilot · JST-GH	1	$89.00	$89.00	Safety FCC. The Pixhawk 6C Mini is the current recommended version. Includes power module PM06. Buy the combo set if available. https://www.amazon.com/dp/B0FBW3RZ9V

Subtotal: $104.00

CAN-FD INTERFACE

Item	Spec	Qty	Unit	Line	Notes / Amazon Link
MCP2518FD SPI-to-CAN-FD module	MCP2518FD breakout · SPI · 5V · CAN-FD 8Mbit/s	1	$12.99	$12.99	Adds CAN-FD to RPi Zero 2W. Search 'MCP2518FD breakout board' or 'CANFD SPI module'. Also available from SparkFun (DEV-20565). Required — RPi has no native CAN-FD. https://www.amazon.com/s?k=MCP2518FD+CAN+FD+breakout
CAN bus termination resistors 120Ω	120Ω · 0.25W · through-hole · 1% tolerance	2	$0.10	$0.20	One at each end of CAN bus. Critical — without these CAN is unreliable. Buy a resistor kit (under $10) and you'll have what you need. https://www.amazon.com/s?k=120+ohm+resistor+through+hole
Twisted pair signal cable 22AWG	22AWG · twisted pair · ~2m needed for CAN runs	1	$4.99	$4.99	For CAN bus wiring. Can also repurpose unused pairs from a CAT5 ethernet cable. https://www.amazon.com/s?k=22awg+twisted+pair+wire

Subtotal: $18.18

POWER SYSTEM

Item	Spec	Qty	Unit	Line	Notes / Amazon Link
6S LiPo 5000mAh 50C XT60	22.2V · 5000mAh · 50C · 6S · XT60 plug	1	$64.99	$64.99	Confirmed Rev H. Search 'Zeee 6S 5000mAh' or 'OVONIC 6S 5000mAh' for good value. ALWAYS store in a LiPo bag — buy one at the same time. https://www.amazon.com/s?k=6s+5000mah+50c+lipo+XT60
LiPo safe storage bag	Fireproof LiPo bag · fits 6S 5000mAh · 23×30cm or larger	1	$9.99	$9.99	Essential safety item. LiPo fires are serious. Buy before the battery. https://www.amazon.com/s?k=lipo+battery+bag+fireproof
XT60 male+female connector pairs	XT60 · gold plated · 5 pairs	1	$6.99	$6.99	Main battery connectors. Buy a pack — you'll use several. https://www.amazon.com/s?k=XT60+connector+pairs
5V 3A BEC / voltage regulator	Input 5–36V · output 5V 3A · switching · small form factor	1	$8.99	$8.99	Powers RPi + all 16 servos from 22V main battery. Search 'Matek UBEC 5V 3A' or 'Pololu D24V22F5'. Must be 6S (22V) input compatible. https://www.amazon.com/s?k=5V+3A+BEC+UBEC+drone
Power module with current sensor	0–60A · 6S · XT60 in/out · JST-GH to Pixhawk · 5.3V BEC	1	$14.99	$14.99	Sits between battery and distribution. Search 'Holybro PM06' or 'FPVDrone power module 6S'. Included with some Pixhawk 6C Mini sets — check before buying separately. https://www.amazon.com/s?k=pixhawk+power+module+6S+current+sensor
Power distribution board	6S · XT60 input · 8× outputs · current rated	1	$11.99	$11.99	Splits battery to 8 ODrives. Search 'Matek FCHUB-6S' or use thick copper bus bars soldered directly. https://www.amazon.com/s?k=power+distribution+board+6S+drone
Capacitors 470µF 50V electrolytic (pack)	470µF · 50V · low ESR · electrolytic · 2 required	1	$6.99	$6.99	Mount close to ODrives. Smooths voltage spikes on motor start/stop. Buy a pack. https://www.amazon.com/s?k=470uf+50v+capacitor+electrolytic+low+ESR

Subtotal: $124.93

SENSORS

Item	Spec	Qty	Unit	Line	Notes / Amazon Link
Matek ASPD-4525 Airspeed Sensor + Pitot	MS4525DO · I2C · 5V · includes pitot tube + tubing	1	$19.99	$19.99	Airspeed sensor for stall protection. Plugs into Pixhawk I2C port. Includes pitot tube and silicon tubing. Also available from getfpv.com and readymaderc.com. https://www.amazon.com/dp/B0D9H5M9NQ
TF-Luna LiDAR Rangefinder 0.2–8m	TF-Luna · 0.2–8m · ±2cm · UART/I2C · 5g · 5V	1	$22.99	$22.99	Ground proximity sensor — controls wing amplitude at takeoff/landing. Points straight down from undercarriage. Plugs into Pixhawk UART or RPi. https://www.amazon.com/dp/B088NVX2L7
Holybro M9N GPS Module with Compass	u-blox M9N · GPS+GLONASS · IST8310 compass · LED · Pixhawk connector	1	$44.99	$44.99	GPS for position hold and return-to-home. Mount on short mast away from motors. M9N is the current recommended module. Some Pixhawk combo sets include GPS — check first. https://www.amazon.com/dp/B0GCZ182YR

Subtotal: $87.97

RC SYSTEM

Item	Spec	Qty	Unit	Line	Notes / Amazon Link
RadioMaster RP3 ELRS Receiver	RP3 · ELRS 2.4GHz · SBUS/CRSF · 5 channels min	1	$14.99	$14.99	Receives RC from transmitter. SBUS output to Pixhawk. Must match TX protocol. RP3 works with RadioMaster TX16S ELRS. If using different TX, buy matching receiver. https://www.amazon.com/s?k=RadioMaster+RP3+ELRS+receiver
RadioMaster TX16S MK II Transmitter	TX16S MKII · ELRS · 16ch · Hall gimbals · EdgeTX · 4.3" display	1	$159.99	$159.99	Your hand controller. Listed for budget completeness — you may own one already. TX16S is the best value full-featured transmitter. Buy ELRS version for long range. https://www.amazon.com/dp/B0B45MDPCW

Subtotal: $174.98

TELEMETRY

Item	Spec	Qty	Unit	Line	Notes / Amazon Link
SiK Telemetry Radio 433MHz pair	SiK 433MHz · 100mW · USB ground station + aircraft module	1	$34.99	$34.99	Wireless MAVLink to laptop during flight. Connects to Pixhawk TELEM1. Search 'Holybro SiK telemetry 433' or '3DR telemetry 433MHz'. Recommended for first flights. https://www.amazon.com/s?k=sik+telemetry+radio+433+pixhawk

Subtotal: $34.99

WIRING AND CONNECTORS

Item	Spec	Qty	Unit	Line	Notes / Amazon Link
Servo extension cable set 30cm (20 pack)	26AWG · JR/Futaba 3-pin · 30cm · male to female	1	$11.99	$11.99	16 needed for servos + 4 spares. Buy a bulk pack. https://www.amazon.com/s?k=servo+extension+cable+30cm+pack
PCA9685 16-channel PWM servo driver	PCA9685 · I2C · 16ch · 5V · servo PWM output	1	$5.99	$5.99	Recommended — drives all 16 servos via I2C from RPi. More reliable than direct GPIO PWM at 50Hz for 16 channels. https://www.amazon.com/s?k=PCA9685+servo+driver+board
Silicone wire 14AWG red+black (3ft each)	Silicone · 14AWG · 3ft red + 3ft black	1	$8.99	$8.99	Main power runs battery→distribution→ODrives. Must be silicone (not PVC) — PVC cracks with vibration and heat. https://www.amazon.com/s?k=14awg+silicone+wire+red+black
Silicone wire 22AWG multicolor	Silicone · 22AWG · assorted colors · ~20ft	1	$7.99	$7.99	Signal wiring: CAN bus, I2C, UART between boards. https://www.amazon.com/s?k=22awg+silicone+wire+kit+multicolor
JST-GH connector kit	JST-GH 1.25mm · 4-pin & 6-pin · housings + crimp pins	1	$12.99	$12.99	All Pixhawk ports use JST-GH. Buy a full kit — you will make many cables. Needs a JST crimping tool ($15–25) if you don't own one. https://www.amazon.com/s?k=JST-GH+connector+kit+1.25mm
JST crimp tool	Micro connector crimping tool · suits JST-GH 1.25mm	1	$18.99	$18.99	Required for making JST-GH cables. Buy once, use forever. Search 'Engineer PA-09 micro connector tool' for quality option. https://www.amazon.com/s?k=JST+GH+crimping+tool+1.25mm
Heat shrink tubing assortment	2mm · 4mm · 6mm · black · adhesive lined · 200+ pieces	1	$8.99	$8.99	Insulation on all soldered joints. Get adhesive-lined for power connections. https://www.amazon.com/s?k=heat+shrink+tubing+assortment+black

Subtotal: $75.93

MOUNTING HARDWARE

Item	Spec	Qty	Unit	Line	Notes / Amazon Link
Anti-vibration mounts M3 (Pixhawk)	M3 silicone anti-vibration standoffs · 4 pack	1	$5.99	$5.99	Critical — Pixhawk IMU is ruined by vibration. Use these under the Pixhawk. Often included with Pixhawk purchase — check before buying separately. https://www.amazon.com/s?k=M3+anti+vibration+mount+flight+controller
Nylon M2/M3 standoff kit	M2 and M3 nylon standoffs + screws · assorted lengths	1	$7.99	$7.99	Board mounting for RPi, PCA9685, MCP2518FD. Nylon prevents shorts. https://www.amazon.com/s?k=nylon+standoff+kit+M2+M3+assorted
SanDisk 32GB MicroSD card	MicroSD · 32GB · Class 10 · A1	1	$8.99	$8.99	RPi boot drive. Install Raspberry Pi OS Lite + firmware files. https://www.amazon.com/s?k=sandisk+32gb+microsd+class10
Short USB-C cable 20cm	USB-C · 20cm · male to male · data + power	1	$5.99	$5.99	Connects 5V BEC to RPi Zero 2W power input. https://www.amazon.com/s?k=USB-C+cable+20cm+short

Subtotal: $28.96

Cost Summary

Category	Subtotal
Flap drive (motors, ODrives, pulleys, rope)	$358.07
Servos (sweep ×8 + pitch ×8)	$175.84
Computers (RPi Zero 2W + Pixhawk 6C Mini)	$104.00
CAN-FD interface	$18.08
Power system (battery, BEC, distribution)	$124.92
Sensors (airspeed, rangefinder, GPS)	$87.97
RC system (receiver + transmitter)	$174.98
Telemetry radio	$34.99
Wiring and connectors	$81.91
Mounting hardware	$37.95
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Total excl. RC transmitter	**~$1,050**
RC transmitter (RadioMaster TX16S MK II)	**$160**
GRAND TOTAL	**~$1,210**

Note on ODrive cost: 8× ODrive Micro at $79 each = $632 — the single largest expense at 52% of the drone electronics cost (excl. transmitter). There is no cheaper substitute that provides equivalent CAN-FD closed-loop velocity control at 6S voltage. This is the correct and necessary component.

MCO-1D Electronics BOM — USD Edition · 2026-06-03 · All Amazon links are search or product pages — verify spec before ordering. ODrive Micro: odriverobotics.com only. Prices approximate. Not included: airframe, wings, tools, LiPo charger (~$35), LiPo safe bag ($10).